(Jephian) Chin-Hung Lin 林晉宏

Phone: (+1)307-314-9275
Email: `jephianlin@nycu.edu.tw`
`https://jephianlin.github.io/`

Department of Applied Mathematics
National Yang Ming Chiao Tung University (NYCU)
Hsinchu 300093, Taiwan

EMPLOYMENT

2026–present Associate professor, NYCU

2022–2026 Associate professor, National Sun Yat-sen University (NSYSU)

2018–2022 Assistant professor, NSYSU

2017–2018 Post-doctoral fellow, University of Victoria (UVic)

EDUCATION

2017 Ph.D., Mathematics, Iowa State University (ISU)

2011 M.S., Mathematics, National Taiwan University (NTU)

2009 B.S., Mathematics, National Taiwan Normal University (NTNU)

RESEARCH INTERESTS

Spectral graph theory; inverse eigenvalue problem; network analysis; quantum information.

HONORS

2026 Plenary Speaker (LAMA Lecture) at the 27th Conference of the International Linear Algebra Society, Blacksburg, VA

2019–2024 Young Scholar Fellowship (愛因斯坦培植計畫), Ministry of Science and Technology, Taiwan (於 2021 年獲選轉爲 2030 跨世代新秀學者計畫)

2022 LAA Early Career Speaker at the 24th Conference of the International Linear Algebra Society, Galway, Ireland

2017 Zaffarano Prize for graduate student research, ISU

PROFESSIONAL SERVICES

Member, Board of Directors of the International Linear Algebra Society (ILAS), 2025–present.

Guest Editor of *Linear Algebra and its Applications (ILAS2025 special issue)*, 2025–2026.

Editor of *Linear and Multilinear Algebra*, 2024–present.

Associate Editor of the *Electronic Journal of Linear Algebra*, 2023–present.

Referee for *Linear Algebra and its Applications*, *Journal of Combinatorial Optimization*, *Discrete Optimization*, *Special Matrices*, and *Discrete Applied Mathematics*, etc.

Local Organizer (Chair) of the 26th Conference of ILAS, 2025.

Assistant Conference Coordinator of the 21th Conference of ILAS, 2017.

PUBLICATIONS

Appeared/Accepted

36. J. C.-H. Lin and M. N. Shirazi. Inverse Fiedler vector problem of a graph. *Linear Algebra Appl.*, 748:80–103, 2026.
35. A. Laikhuram and J. C.-H. Lin. Inverse eigenvalue problem for discrete Schrödinger operators of a graph. *Linear Algebra Appl.*, 730:566–586, 2026.
34. S. M. Fallat, H. Gupta, and J. C.-H. Lin. Inverse eigenvalue problem for Laplacian matrices of a graph. *SIAM J. Matrix Anal. Appl.*, 46:1866–1886, 2025.
33. A. Abiad, B. A. Curtis, M. Flagg, H. T. Hall, J. C.-H. Lin, and B. Shader. The inverse nullity pair problem and the strong nullity interlacing property. *Linear Algebra Appl.*, 699:539–568, 2024.
32. C. Erickson, L. Gan, J. Kritschgau, J. C.-H. Lin, and S. Spiro. Complementary vanishing graphs. *Linear Algebra Appl.*, 692:185–211, 2024.
31. P. Hell, J. Huang, and J. C.-H. Lin. Strong cocomparability graphs and slash-free orderings of matrices. *SIAM J. Discrete Math.*, 38:828–844, 2024.
30. J. C.-H. Lin, P. Oblak, and H. Šmigoc. The liberation set in the inverse eigenvalue problem of a graph. *Linear Algebra Appl.*, 675:1–28, 2023.
29. S. M. Fallat, H. T. Hall, J. C.-H. Lin, and B. Shader. The bifurcation lemma for strong properties in the inverse eigenvalue problem of a graph. *Linear Algebra Appl.*, 648:70–87, 2022.
28. J. C.-H. Lin, P. Oblak, and H. Šmigoc. On the inverse eigenvalue problem for block graphs. *Linear Algebra Appl.*, 631:379–397, 2021.
27. F. H. J. Kenter and J. C.-H. Lin. A zero forcing technique for bounding sums of eigenvalue multiplicities. *Linear Algebra Appl.*, 629:138–167, 2021.
26. P. Hell, C. Hernandez-Cruz, J. Huang, and J. C.-H. Lin. Strong chordality of graphs with possible loops. *SIAM J. Discrete Math.*, 35:362–375, 2021.
25. L. Hogben, J. C.-H. Lin, D. D. Olesky, and P. van den Driessche. The sepr-sets of sign patterns. *Linear Multilinear Algebra*, 26:2044–2068, 2020.
24. P. Hell, J. Huang, J. C.-H. Lin, and R. M. McConnell. Bipartite analogues of comparability and cocomparability graphs. *SIAM J. Discrete Math.*, 34:1969–1983, 2020.
23. S. Butler, C. Erickson, S. M. Fallat, H. T. Hall, B. Kroschel, J. C.-H. Lin, B. Shader, N. Warnberg, and B. Yang. Properties of a q-analogue of zero forcing. *Graphs Combin.*, 36:1401–1419, 2020.
22. A. Chan, S. M. Fallat, S. Kirkland, J. C.-H. Lin, S. Nasserasr, and S. Plosker. Complex Hadamard diagonalisable graphs. *Linear Algebra Appl.*, 605:158–179, 2020.
21. J. C.-H. Lin, P. Oblak, and H. Šmigoc. The strong spectral property for graphs. *Linear Algebra Appl.*, 598:68–91, 2020.
20. W. Barrett, S. Butler, S. M. Fallat, H. T. Hall, L. Hogben, J. C.-H. Lin, B. Shader, and M. Young. The inverse eigenvalue problem of a graph: Multiplicities and minors. *J. Combin. Theory Ser. B*, 142:276–306, 2020.

19. D. Ferrero, M. Flagg, H. T. Hall, L. Hogben, J. C.-H. Lin, S. Meyer, S. Nasserasr, and B. Shader. Rigid linkages and partial zero forcing. *Electron. J. Combin.*, 26:#P2.43, 2019.
18. F. H. J. Kenter and J. C.-H. Lin. On the error of a priori sampling: Zero forcing sets and propagation time. *Linear Algebra Appl.*, 576:124–141, 2019.
17. C. A. Alfaro and J. C.-H. Lin. Critical ideals, minimum rank and zero forcing number. *Appl. Math. Comput.*, 358:305–313, 2019.
16. J. C.-H. Lin. Zero forcing number, Grundy domination number, and their variants. *Linear Algebra Appl.*, 563:240–254, 2019.
15. Y.-J. Cheng and J. C.-H. Lin. Graph families with constant distance determinant. *Electron. J. Combin.*, 25:#P4.45, 2018.
14. R. Anderson, S. Bai, F. Barrera-Cruz, É. Czabarka, G. Da Lozzo, N. L. F. Hobson, J. C.-H. Lin, A. Mohr, H. C. Smith, L. A. Székely, and H. Whitlatch. Analogies between the crossing number and the tangle crossing number. *Electron. J. Combin.*, 25:#P4.24, 2018.
13. G. Aalipour, A. Abiad, Z. Berikkyzy, L. Hogben, F. H. J. Kenter, J. C.-H. Lin, and M. Tait. Proof of a conjecture of Graham and Lovász concerning unimodality of coefficients of the distance characteristic polynomial of a tree. *Electron. J. Linear Algebra*, 34:373–380, 2018.
12. J. C.-H. Lin, D. D. Olesky, and P. van den Driessche. Sign patterns requiring a unique inertia. *Linear Algebra Appl.*, 546:67–85, 2018.
11. W. Barrett, S. M. Fallat, H. T. Hall, L. Hogben, J. C.-H. Lin, and B. Shader. Generalizations of the Strong Arnold Property and the minimum number of distinct eigenvalues of a graph. *Electron. J. Combin.*, 24:#P2.40, 2017.
10. A. Berliner, C. Bozeman, S. Butler, M. Catral, L. Hogben, B. Kroschel, J. C.-H. Lin, N. Warnberg, and M. Young. Zero forcing propagation time on oriented graphs. *Discrete Appl. Math.*, 224:45–59, 2017.
9. M. Dairyko, L. Hogben, J. C.-H. Lin, J. Lockhart, D. Roberson, S. Severini, and M. Young. Note on von Neumann and Rényi entropies of a graph. *Linear Algebra Appl.*, 521:240–253, 2017.
8. J. C.-H. Lin. Using a new zero forcing process to guarantee the Strong Arnold Property. *Linear Algebra Appl.*, 507:229–250, 2016.
7. S. Butler, C. Erickson, L. Hogben, K. Hogenson, L. Kramer, R. L. Kramer, J. C.-H. Lin, R. R. Martin, D. Stolee, N. Warnberg, and M. Young. Rainbow arithmetic progressions. *J. Comb.*, 7:595–626, 2016.
6. G. Aalipour, A. Abiad, Z. Berikkyzy, J. Cummings, J. De Silva, W. Gao, K. Heysse, L. Hogben, F. H. J. Kenter, J. C.-H. Lin, and M. Tait. On the distance spectra of graphs. *Linear Algebra Appl.*, 497:66–87, 2016.
5. J. C.-H. Lin. Odd cycle zero forcing parameters and the minimum rank of graph blowups. *Electron. J. Linear Algebra*, 31:42–59, 2016.
4. C. Bozeman, A. Ellsworth, L. Hogben, J. C.-H. Lin, G. Maurer, K. Nowak, A. Rodriguez, and J. Strickland. Minimum rank of graphs with loops. *Electron. J. Linear Algebra*, 27:907–934, 2014.

3. J. C.-H. Lin. The sieving process and lower bounds for the minimum rank problem. *Congr. Numer.*, 219:73–88, 2014.
2. G. J. Chang and J. C.-H. Lin. Counterexamples to an edge spread question for zero forcing number. *Linear Algebra Appl.*, 446:192–195, 2014.
1. J. C.-H. Lin. Some interpretations and applications of Fuss-Catalan numbers. *ISRN Discrete Math.*, 2011. doi:10.5402/2011/534628.

Book/Others

3. L. Hogben, J. C.-H. Lin, and B. Shader. *Inverse Problems and Zero Forcing for Graphs*. American Mathematical Society, Providence, 2022.
2. S. M. Fallat, L. Hogben, J. C.-H. Lin, and B. Shader. The inverse eigenvalue problem of a graph, zero forcing, and related parameters. *Notices Amer. Math. Soc.*, 67:257–261, February, 2020.
1. L. Hogben, J. C.-H. Lin, and B. Shader. The inverse eigenvalue problem of a graph. In *50 Years of Combinatorics, Graph Theory, and Computing*, 1st edition, F. Chung, R. Graham, F. Hoffman, L. Hogben, R. C. Mullin, and D. B. West editors, CRC Press, Boca Raton, 2019.

SELECTED PRESENTATIONS

Plenary Lectures

2026 "Inverse problems on a graph: strong matrices and graph minors," 27th International Linear Algebra Society Conference, Blacksburg, VA (LAMA Lecture).

2025 "Inverse problems on a graph: strong matrices and graph minors," International Conference in Discrete Mathematics, Tiruchirappalli, India.

2017 "Variants of Zero Forcing," AIM Workshop: Zero forcing and its applications, San Jose, CA.

Invited for Special Sessions/Mini Symposia

2026 "Jacobian method and strong properties," 27th International Linear Algebra Society Conference, Blacksburg, VA.

2026 "Inverse Fiedler vector problem of a graph," The 10th Young Scholar Symposium of the East Asia Section of IPIA, Taipei, Taiwan.

2025 "Inverse Fiedler vector problem of a graph," 26th International Linear Algebra Society Conference, Kaohsiung, Taiwan.

2025 "Strong properties from a universal point of view," Joint Mathematics Meetings, Seattle, WA.

2024 "The liberation set in the inverse eigenvalue problem of a graph," Workshop on Matrices and Operators, Reno, NV.

2023 "The bifurcation lemma for strong properties in the inverse eigenvalue problem of a graph," 25th International Linear Algebra Society Conference, Madrid, Spain.

2023 "The bifurcation lemma for strong properties in the inverse eigenvalue problem of a graph," Joint Mathematics Meetings, Boston, MA.

2022 "Comparability and cocomparability bigraphs," 24th International Linear Algebra Society Conference, Galway, Ireland (LAA early career speaker).

2022 "On the inverse eigenvalue problem for block graphs," Joint Mathematics Meetings, Seattle, WA.

2021 "The strong spectral property for graphs," SIAM Conference on Applied Linear Algebra with the embedded 23rd International Linear Algebra Society Conference, virtual.

2021 "The strong spectral property for graphs," Canadian Discrete and Algorithmic Mathematics Conference, virtual.

2021 "The strong spectral property for graphs," Joint Mathematics Meetings, virtual.

2019 "Zero forcing number, Grundy domination number and their variants," 22th International Linear Algebra Society Conference, Rio de Janeiro, Brazil.

2018 "Graphs whose distance matrices have the same determinant," SIAM Conference on Discrete Mathematics, Denver, CO.

2018 "The inverse eigenvalue problem of a graph: Multiplicities and minors," Joint Mathematics Meetings, San Diego, CA.

2017 "General spectral graph theory: The inverse eigenvalue problem of a graph," Combinatorial Potlatch, Victoria, BC, Canada.

2017 "Note on von Neumann and Rényi entropies of a graph," 21th International Linear Algebra Society Conference, Ames, IA.

2016 "Distance Spectra of Graphs," AMS Fall Central Sectional Meeting, Minneapolis, MN.

2016 "Using a new zero forcing process to guarantee the Strong Arnold Property," 20th International Linear Algebra Society Conference, Leuven, Belgium.

2016 "Using a new zero forcing process to guarantee the Strong Arnold Property," AMS Spring Central Sectional Meeting, Fargo, ND.

2016 "Odd cycle zero forcing parameters and the minimum rank problem," 47th Southeastern International Conference on Combinatorics, Graph Theory, and Computing, Boca Raton, FL.

2014 "Reduction identities of the minimum rank on loop graphs," 19th International Linear Algebra Society Conference (Satellite Conference of International Congress of Mathematicians 2014), Seoul, S. Korea.